Deep Q-Learning for Lane Localization

Exploring Reinforcement Learning for Accurate Lane Detection

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Outline

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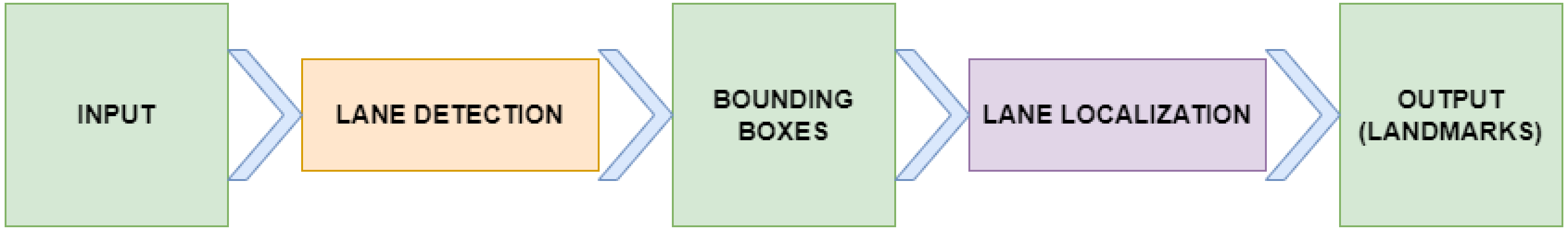
Driving Force: Why This Matters

- Lane keeping is backbone of autonomous driving
- Struggles with different lighting, weather, occlusions
- Lane detection
  - identifies the presence and location of lanes.
- Lane Localization
  - refine the detected lanes

Baseline

- In the paper 'Deep reinforcement learning based lane detection and localization' by Zhao et al. using **Deep Q-learning** for localization is proposed



Simplified flowchart of lane detection and localization framework in paper by Zhao et al.

Research Question

To explore Deep Q-Learning (DQL) with **curve** manipulation

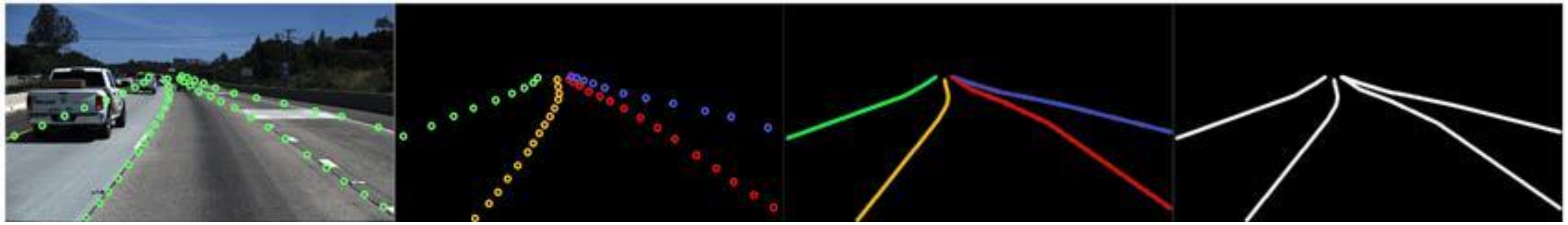
1. Which types of curves are best suited?

2. How can curves be adjusted?

3. Which distance metrics are most suitable?

Data

- **Datasets:** Employed the TuSimple dataset, comprising 6408 road images from US highways for training, validation, and testing



Sample Image and labelled annotations from TuSimple

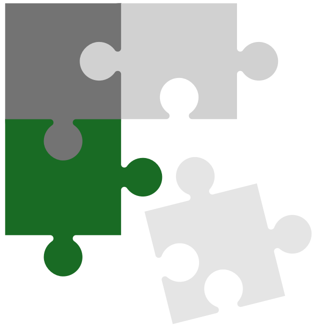
Baseline Outcome

- HitRate = Correct points / All points
- HitRate – 0.65 vs Detect HitRate – 0.731

Results from Unmodified Base Implementation

Baseline Issues

- Rescaling artifacts
  - Fix: Intelligent cropping to minimize distortion
- Misconfigured Image Normalization
  - Fix: Corrected data preprocessing
- Limited Lane Position Representation
  - Fix: Expanded lane representation from 5 to 50 positions



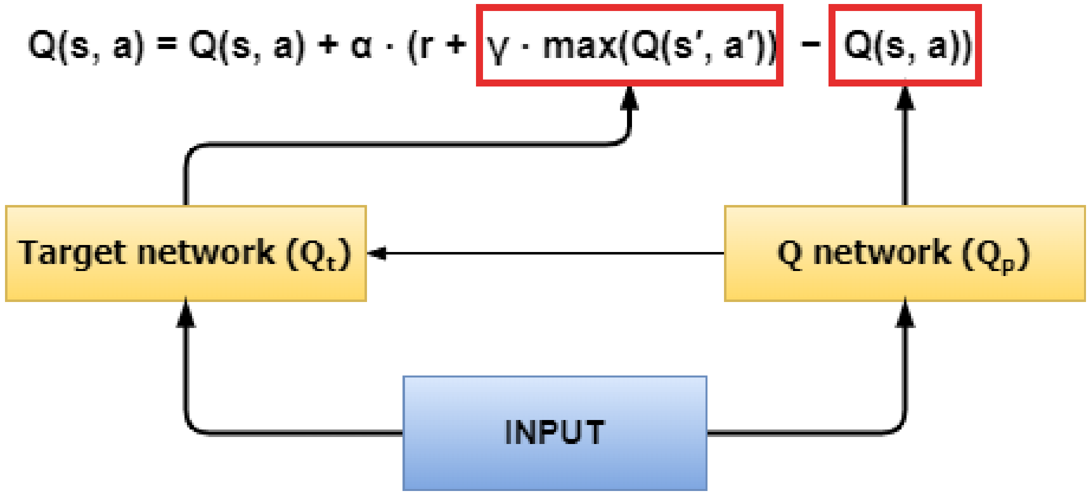
Baseline Issues

- Dependency on Inbuilt Feature Extractor
  - Fix: Used pre-trained ResNet-50 feature extractor, to focus on RL task
- Lack of Exploration
  - Fix: Adopted an epsilon-greedy policy for exploration
- Flawed loss Implementation and Bellman’s equation
  - Fix: Corrected the equations

Enhancements

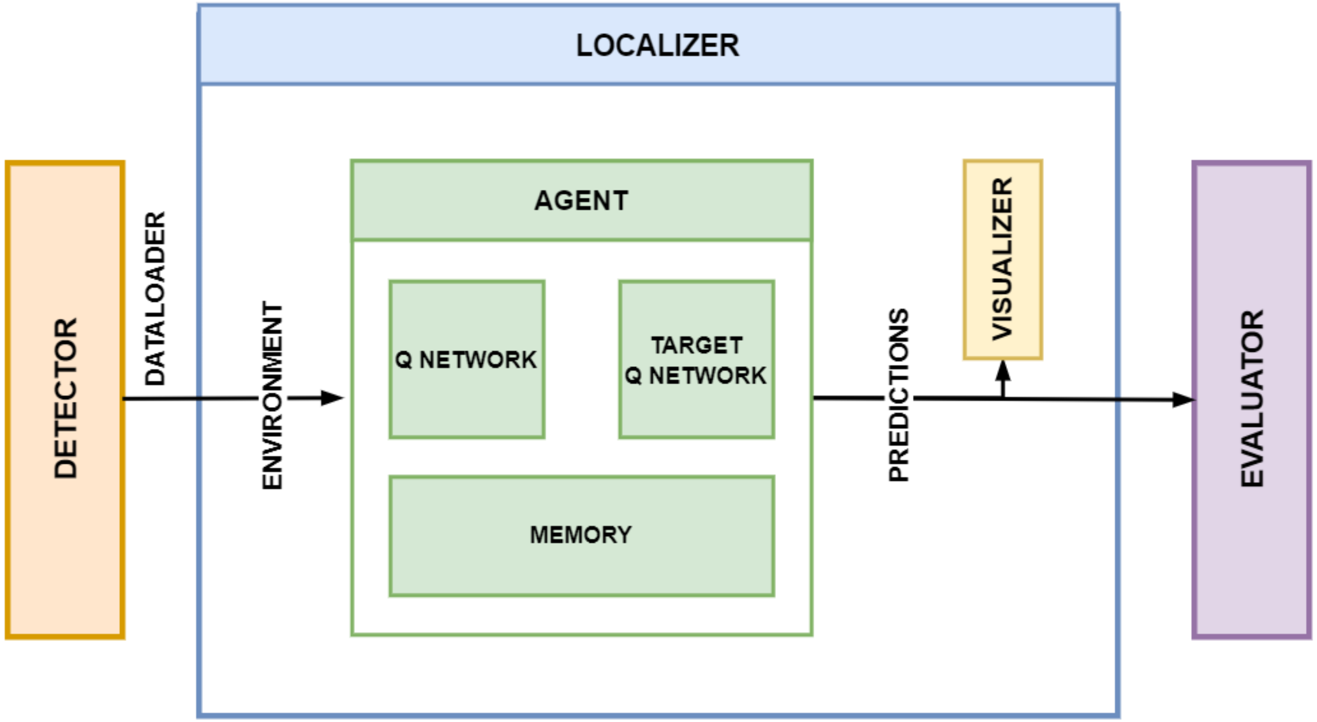
Enhancing DQN with Double DQN (D-DQN):

- Policy Q-Network:
  - For action selection, updated at every step
- Target Q-Network:
  - For value estimation, updated intermittently, enhancing stability and accuracy



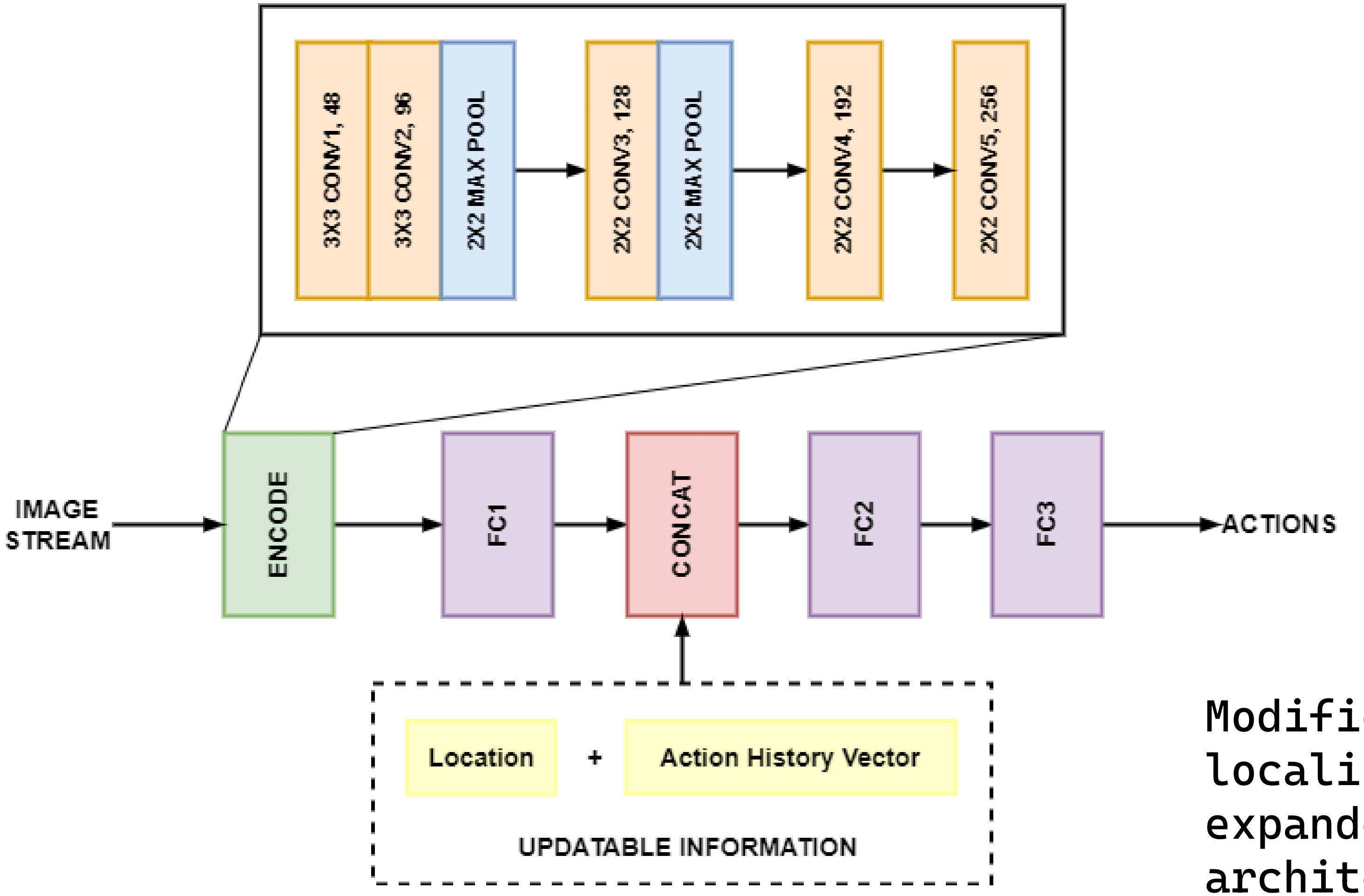
Q-Learning in D-DQN

Architecture



Software architecture of experimental setup

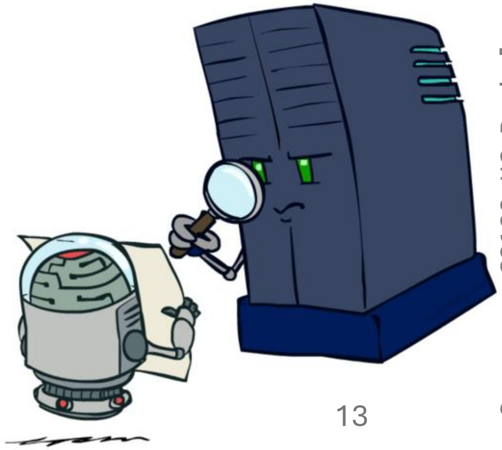
Agent Architecture



Modified model architecture of localizer using points with expanded custom encoder architecture

Evaluation

- HitRate (at cropped image level)
  - Hitrate = Correct points / All points
- Accuracy
  - acc = Correct Predictions / Total Lane Points
- False Positive Rate (FPR)
  - FPR = Incorrect Predictions / Actual Non-Lane Points
- False Negative Rate (FNR)
  - FNR = Missed Lanes / Actual Lanes



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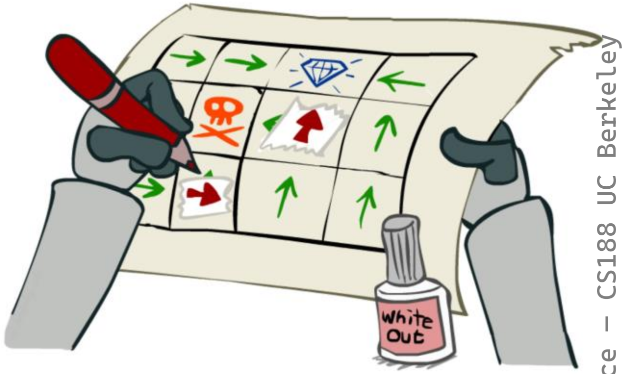
Iterative Modification Results

	Hitrate
Modified Detector (without localizer)	0.784
Using entire lane image	0.469
D-DQN with MSE	0.298
D-DQN with SmoothLoss	0.545
D-DQN with SmoothLoss + priority replay	0.609
D-DQN with SmoothLoss + priority replay + hard update	0.463
D-DQN with SmoothLoss + priority replay + soft update	**0.614**

Curve Manipulation



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- Increased load from using all lane points versus a simplified curve approach

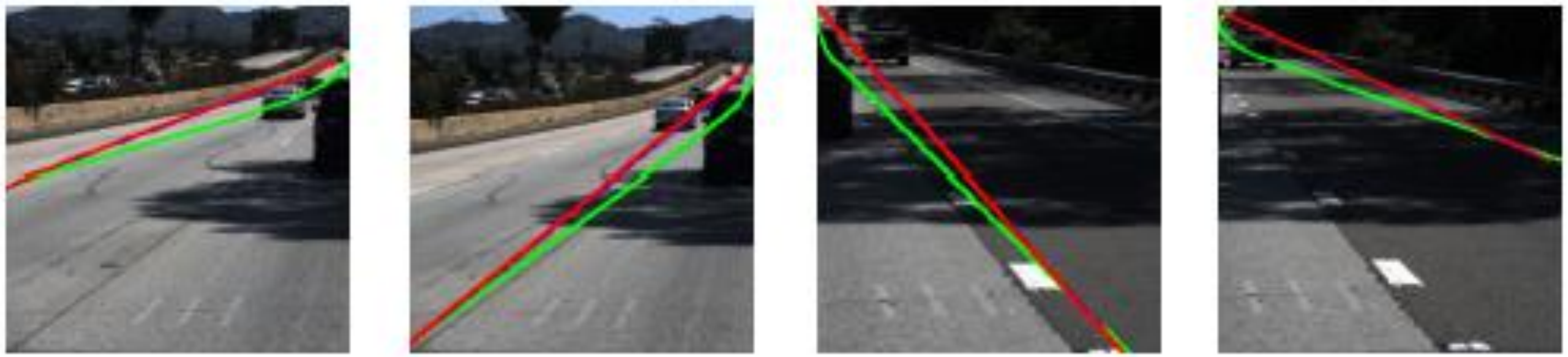
- Bézier Curves:
  - Simpler than Polynomial, and less parameters than B-Spline
  - 4-point Bézier's action space much larger 3-point Bézier
- Ground Truth & Distance Metrics:
  - Ground truth is kept as lane points instead of converted curves
  - So, no curve-based distance metric needed

Evaluation of Localizers

	Hitrate	acc	FPR	FNR
Detector	0.784	0.900	0.233	0.220
Point Localizer (custom extractor)	0.610	0.805	0.449	0.422
Point Localizer (ResNet-50 extractor, Hist=0)	0.626	0.892	0.239	0.226
Point Localizer (ResNet-50 extractor, Hist=8)	0.612	0.888	0.245	0.233
Beizer Curve Localizer	0.590	0.783	0.452	0.446

Visualization of Localizers

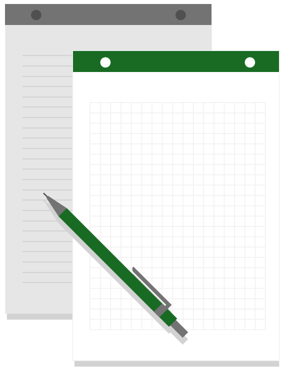
Lane visualization of Point Localizer with pre-trained ResNet-50 feature extractor and no history frames



Lane visualization of Curve Localizer with pre-trained ResNet-50 feature extractor and no history frames

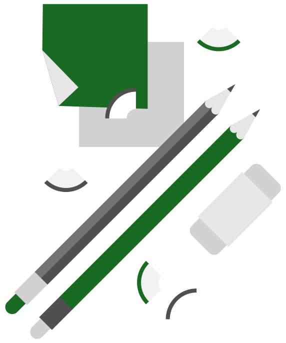
Discussion

- Localizer worse than initial detection
  - 0.892 vs 0.900
- Curves do not provide any advantage over point localizer
  - End to End Curve: BézierLaneNet – 0.9565



Discussion

- Agent did not see all images – even after 100k steps
  - Point – number of lane points + actions per point
  - Curve – actions per lane
- History vector – actions vs previous state – no normalization
- Curves – ended as straight lines – not easy to manipulate



Conclusions

- Proposed curve manipulation failed due to modelling complexity
- Flawed base implementation, and the required fixes resulted in deviating from the initial plan
- Obtained results offered limited interpretability

Source – CS188 UC Berkeley

Conclusions

- Enhancements on DQN to D-DQN and prioritized replay did improve performance and learning

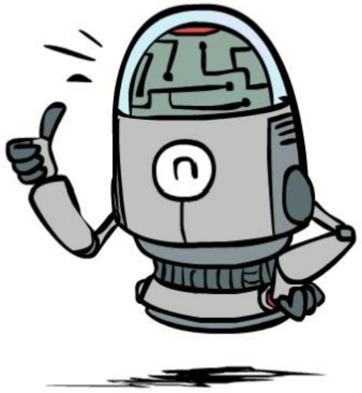
- DQL is not suitable for lane localization in its current state

- Use standard benchmarking metrics to see actual performance

Source – CS188 UC Berkeley

Thank you.

Questions?



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